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1 D:\anaconda3\envs\graduate\python.exe "D:\Temporary
  files\graduateDesign\github\LSTM-Music-Genre-
  Classification-master - 2\
  lstm_genre_classifier_pytorch.py"
2 D:\anaconda3\envs\graduate\lib\site-packages\paramiko
  \transport.py:219: CryptographyDeprecationWarning:
  Blowfish has been deprecated
3   "class": algorithms.Blowfish,
4 Preprocessed files exist, deserializing npy files
5 Training X shape: (2100, 128, 33)
6 Training Y shape: (2100, 5)
7 Validation X shape: (900, 128, 33)
8 Validation Y shape: (900, 5)
9 Test X shape: (900, 128, 33)
10 Test Y shape: (900, 5)
11 Build LSTM RNN model ...
12
13 Training on GPU
14 Training ...
15 Epoch:  0 | NLLoss: 0.9072 | Train Accuracy: 62.19
16 Validation ...
17 Epoch:  0 | NLLoss: 0.9072 | Train Accuracy: 62.19 |
  Val Loss 1.9681 | Val Accuracy: 42.97
18 Epoch:  1 | NLLoss: 0.8579 | Train Accuracy: 63.95
19 Epoch:  2 | NLLoss: 0.8583 | Train Accuracy: 65.29
20 Epoch:  3 | NLLoss: 0.8742 | Train Accuracy: 64.14
21 Epoch:  4 | NLLoss: 0.8360 | Train Accuracy: 66.14
22 Epoch:  5 | NLLoss: 0.8859 | Train Accuracy: 63.38
23 Epoch:  6 | NLLoss: 0.9287 | Train Accuracy: 61.52
24 Epoch:  7 | NLLoss: 0.8916 | Train Accuracy: 64.14
25 Epoch:  8 | NLLoss: 0.8827 | Train Accuracy: 65.00
26 Epoch:  9 | NLLoss: 0.8385 | Train Accuracy: 65.14
27 Epoch: 10 | NLLoss: 0.8429 | Train Accuracy: 66.00
28 Validation ...
29 Epoch: 10 | NLLoss: 0.8429 | Train Accuracy: 66.00
  | Val Loss 1.9266 | Val Accuracy: 41.83
30 Epoch: 11 | NLLoss: 0.8775 | Train Accuracy: 64.10
31 Epoch: 12 | NLLoss: 0.8731 | Train Accuracy: 64.19
32 Epoch: 13 | NLLoss: 0.8932 | Train Accuracy: 63.38
33 Epoch: 14 | NLLoss: 0.8542 | Train Accuracy: 64.81
34 Epoch: 15 | NLLoss: 0.8006 | Train Accuracy: 67.48
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35 Epoch: 16 | NLLoss: 0.7842 | Train Accuracy: 68.52
36 Epoch: 17 | NLLoss: 0.8026 | Train Accuracy: 67.10
37 Epoch: 18 | NLLoss: 0.8146 | Train Accuracy: 66.38
38 Epoch: 19 | NLLoss: 0.8388 | Train Accuracy: 65.62
39 Epoch: 20 | NLLoss: 0.9816 | Train Accuracy: 62.00
40 Validation ...
41 Epoch: 20 | NLLoss: 0.9816 | Train Accuracy: 62.00
    | Val Loss 2.1100 | Val Accuracy: 42.63
42 Epoch: 21 | NLLoss: 0.9938 | Train Accuracy: 58.71
43 Epoch: 22 | NLLoss: 0.8715 | Train Accuracy: 64.43
44 Epoch: 23 | NLLoss: 0.8253 | Train Accuracy: 66.62
45 Epoch: 24 | NLLoss: 0.8216 | Train Accuracy: 66.38
46 Epoch: 25 | NLLoss: 0.8405 | Train Accuracy: 65.67
47 Epoch: 26 | NLLoss: 0.7856 | Train Accuracy: 68.38
48 Epoch: 27 | NLLoss: 0.7713 | Train Accuracy: 68.90
49 Epoch: 28 | NLLoss: 0.7451 | Train Accuracy: 70.00
50 Epoch: 29 | NLLoss: 0.7484 | Train Accuracy: 69.38
51 Epoch: 30 | NLLoss: 0.7298 | Train Accuracy: 70.05
52 Validation ...
53 Epoch: 30 | NLLoss: 0.7298 | Train Accuracy: 70.05
    | Val Loss 1.9907 | Val Accuracy: 43.77
54 Epoch: 31 | NLLoss: 0.7220 | Train Accuracy: 70.10
55 Epoch: 32 | NLLoss: 0.7216 | Train Accuracy: 70.81
56 Epoch: 33 | NLLoss: 0.7232 | Train Accuracy: 70.05
57 Epoch: 34 | NLLoss: 0.7097 | Train Accuracy: 70.86
58 Epoch: 35 | NLLoss: 0.7152 | Train Accuracy: 70.43
59 Epoch: 36 | NLLoss: 0.7181 | Train Accuracy: 70.81
60 Epoch: 37 | NLLoss: 0.7196 | Train Accuracy: 70.81
61 Epoch: 38 | NLLoss: 0.8325 | Train Accuracy: 66.52
62 Epoch: 39 | NLLoss: 0.9588 | Train Accuracy: 62.62
63 Epoch: 40 | NLLoss: 0.8840 | Train Accuracy: 63.52
64 Validation ...
65 Epoch: 40 | NLLoss: 0.8840 | Train Accuracy: 63.52
    | Val Loss 1.8673 | Val Accuracy: 45.71
66 Epoch: 41 | NLLoss: 0.8000 | Train Accuracy: 66.67
67 Epoch: 42 | NLLoss: 0.7771 | Train Accuracy: 68.33
68 Epoch: 43 | NLLoss: 0.8477 | Train Accuracy: 65.14
69 Epoch: 44 | NLLoss: 0.7594 | Train Accuracy: 68.81
70 Epoch: 45 | NLLoss: 0.9004 | Train Accuracy: 63.38
71 Epoch: 46 | NLLoss: 0.9329 | Train Accuracy: 63.10
72 Epoch: 47 | NLLoss: 0.8342 | Train Accuracy: 66.33
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73 Epoch: 48 | NLLoss: 0.7741 | Train Accuracy: 68.71
74 Epoch: 49 | NLLoss: 0.8027 | Train Accuracy: 67.29
75 Epoch: 50 | NLLoss: 0.8860 | Train Accuracy: 63.62
76 Validation ...
77 Epoch: 50 | NLLoss: 0.8860 | Train Accuracy: 63.62
   | Val Loss 1.8770 | Val Accuracy: 46.29
78 Epoch: 51 | NLLoss: 0.7556 | Train Accuracy: 69.76
79 Epoch: 52 | NLLoss: 0.7535 | Train Accuracy: 69.71
80 Epoch: 53 | NLLoss: 0.7355 | Train Accuracy: 69.14
81 Epoch: 54 | NLLoss: 0.7291 | Train Accuracy: 69.76
82 Epoch: 55 | NLLoss: 0.6998 | Train Accuracy: 70.86
83 Epoch: 56 | NLLoss: 0.7553 | Train Accuracy: 68.29
84 Epoch: 57 | NLLoss: 0.7061 | Train Accuracy: 70.14
85 Epoch: 58 | NLLoss: 0.6869 | Train Accuracy: 71.76
86 Epoch: 59 | NLLoss: 0.6902 | Train Accuracy: 71.76
87 Epoch: 60 | NLLoss: 0.6910 | Train Accuracy: 71.86
88 Validation ...
89 Epoch: 60 | NLLoss: 0.6910 | Train Accuracy: 71.86
   | Val Loss 1.9633 | Val Accuracy: 46.74
90 Epoch: 61 | NLLoss: 0.8742 | Train Accuracy: 64.33
91 Epoch: 62 | NLLoss: 1.0155 | Train Accuracy: 59.67
92 Epoch: 63 | NLLoss: 0.8986 | Train Accuracy: 62.33
93 Epoch: 64 | NLLoss: 0.8656 | Train Accuracy: 64.76
94 Epoch: 65 | NLLoss: 0.8099 | Train Accuracy: 67.10
95 Epoch: 66 | NLLoss: 0.7708 | Train Accuracy: 68.33
96 Epoch: 67 | NLLoss: 0.7342 | Train Accuracy: 69.71
97 Epoch: 68 | NLLoss: 0.7026 | Train Accuracy: 71.38
98 Epoch: 69 | NLLoss: 0.6838 | Train Accuracy: 71.33
99 Epoch: 70 | NLLoss: 0.6664 | Train Accuracy: 72.05
100 Validation ...
101 Epoch: 70 | NLLoss: 0.6664 | Train Accuracy: 72.05
   | Val Loss 1.8599 | Val Accuracy: 47.09
102 Epoch: 71 | NLLoss: 0.6614 | Train Accuracy: 72.29
103 Epoch: 72 | NLLoss: 0.6364 | Train Accuracy: 73.95
104 Epoch: 73 | NLLoss: 0.6360 | Train Accuracy: 73.95
105 Epoch: 74 | NLLoss: 0.6224 | Train Accuracy: 74.38
106 Epoch: 75 | NLLoss: 0.6411 | Train Accuracy: 73.62
107 Epoch: 76 | NLLoss: 0.7546 | Train Accuracy: 69.48
108 Epoch: 77 | NLLoss: 0.7446 | Train Accuracy: 69.95
109 Epoch: 78 | NLLoss: 0.6486 | Train Accuracy: 72.95
110 Epoch: 79 | NLLoss: 0.6436 | Train Accuracy: 73.10

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111 Epoch: 80 | NLLoss: 0.6335 | Train Accuracy: 73.48
112 Validation ...
113 Epoch: 80 | NLLoss: 0.6335 | Train Accuracy: 73.48
    | Val Loss 1.9533 | Val Accuracy: 48.11
114 Epoch: 81 | NLLoss: 0.6372 | Train Accuracy: 73.90
115 Epoch: 82 | NLLoss: 0.6349 | Train Accuracy: 73.95
116 Epoch: 83 | NLLoss: 0.6418 | Train Accuracy: 74.43
117 Epoch: 84 | NLLoss: 0.6126 | Train Accuracy: 74.71
118 Epoch: 85 | NLLoss: 0.6170 | Train Accuracy: 75.14
119 Epoch: 86 | NLLoss: 0.6193 | Train Accuracy: 74.52
120 Epoch: 87 | NLLoss: 0.6228 | Train Accuracy: 74.86
121 Epoch: 88 | NLLoss: 0.6268 | Train Accuracy: 74.43
122 Epoch: 89 | NLLoss: 0.6163 | Train Accuracy: 75.24
123 Epoch: 90 | NLLoss: 0.6374 | Train Accuracy: 73.71
124 Validation ...
125 Epoch: 90 | NLLoss: 0.6374 | Train Accuracy: 73.71
    | Val Loss 1.9925 | Val Accuracy: 47.31
126 Epoch: 91 | NLLoss: 0.7184 | Train Accuracy: 71.14
127 Epoch: 92 | NLLoss: 0.7719 | Train Accuracy: 68.52
128 Epoch: 93 | NLLoss: 0.7423 | Train Accuracy: 70.48
129 Epoch: 94 | NLLoss: 0.6789 | Train Accuracy: 73.43
130 Epoch: 95 | NLLoss: 0.7152 | Train Accuracy: 71.24
131 Epoch: 96 | NLLoss: 0.7554 | Train Accuracy: 68.95
132 Epoch: 97 | NLLoss: 0.7228 | Train Accuracy: 70.57
133 Epoch: 98 | NLLoss: 0.6470 | Train Accuracy: 74.52
134 Epoch: 99 | NLLoss: 0.6545 | Train Accuracy: 74.14
135 Epoch: 100 | NLLoss: 0.6197 | Train Accuracy: 75.10
136 Validation ...
137 Epoch: 100 | NLLoss: 0.6197 | Train Accuracy: 75.10
    | Val Loss 2.0444 | Val Accuracy: 47.77
138 Epoch: 101 | NLLoss: 0.5899 | Train Accuracy: 76.10
139 Epoch: 102 | NLLoss: 0.5807 | Train Accuracy: 76.43
140 Epoch: 103 | NLLoss: 0.5888 | Train Accuracy: 76.24
141 Epoch: 104 | NLLoss: 0.6123 | Train Accuracy: 75.81
142 Epoch: 105 | NLLoss: 0.6890 | Train Accuracy: 71.57
143 Epoch: 106 | NLLoss: 0.8208 | Train Accuracy: 66.95
144 Epoch: 107 | NLLoss: 0.7168 | Train Accuracy: 70.95
145 Epoch: 108 | NLLoss: 0.7990 | Train Accuracy: 66.76
146 Epoch: 109 | NLLoss: 0.6866 | Train Accuracy: 72.33
147 Epoch: 110 | NLLoss: 0.6516 | Train Accuracy: 74.14
148 Validation ...
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149 Epoch: 110 | NLLoss: 0.6516 | Train Accuracy: 74.14
    | Val Loss 1.9877 | Val Accuracy: 48.34
150 Epoch: 111 | NLLoss: 0.6649 | Train Accuracy: 74.24
151 Epoch: 112 | NLLoss: 0.7046 | Train Accuracy: 73.05
152 Epoch: 113 | NLLoss: 0.7626 | Train Accuracy: 70.38
153 Epoch: 114 | NLLoss: 0.6400 | Train Accuracy: 74.33
154 Epoch: 115 | NLLoss: 0.5959 | Train Accuracy: 76.38
155 Epoch: 116 | NLLoss: 0.5715 | Train Accuracy: 76.76
156 Epoch: 117 | NLLoss: 0.5554 | Train Accuracy: 77.90
157 Epoch: 118 | NLLoss: 0.5465 | Train Accuracy: 78.05
158 Epoch: 119 | NLLoss: 0.5431 | Train Accuracy: 78.43
159 Epoch: 120 | NLLoss: 0.5279 | Train Accuracy: 79.38
160 Validation ...
161 Epoch: 120 | NLLoss: 0.5279 | Train Accuracy: 79.38
    | Val Loss 2.0126 | Val Accuracy: 49.71
162 Epoch: 121 | NLLoss: 0.5253 | Train Accuracy: 79.38
163 Epoch: 122 | NLLoss: 0.5167 | Train Accuracy: 80.05
164 Epoch: 123 | NLLoss: 0.5188 | Train Accuracy: 79.81
165 Epoch: 124 | NLLoss: 0.5158 | Train Accuracy: 80.10
166 Epoch: 125 | NLLoss: 0.5451 | Train Accuracy: 78.19
167 Epoch: 126 | NLLoss: 0.5441 | Train Accuracy: 78.19
168 Epoch: 127 | NLLoss: 0.5380 | Train Accuracy: 79.10
169 Epoch: 128 | NLLoss: 0.5589 | Train Accuracy: 78.38
170 Epoch: 129 | NLLoss: 0.6082 | Train Accuracy: 76.52
171 Epoch: 130 | NLLoss: 0.6635 | Train Accuracy: 73.29
172 Validation ...
173 Epoch: 130 | NLLoss: 0.6635 | Train Accuracy: 73.29
    | Val Loss 2.0661 | Val Accuracy: 48.23
174 Epoch: 131 | NLLoss: 0.6166 | Train Accuracy: 75.67
175 Epoch: 132 | NLLoss: 0.5910 | Train Accuracy: 75.57
176 Epoch: 133 | NLLoss: 0.5664 | Train Accuracy: 77.81
177 Epoch: 134 | NLLoss: 0.6566 | Train Accuracy: 73.24
178 Epoch: 135 | NLLoss: 0.6020 | Train Accuracy: 75.81
179 Epoch: 136 | NLLoss: 0.5627 | Train Accuracy: 76.33
180 Epoch: 137 | NLLoss: 0.5392 | Train Accuracy: 78.71
181 Epoch: 138 | NLLoss: 0.6066 | Train Accuracy: 75.33
182 Epoch: 139 | NLLoss: 0.6026 | Train Accuracy: 76.33
183 Epoch: 140 | NLLoss: 0.5832 | Train Accuracy: 76.76
184 Validation ...
185 Epoch: 140 | NLLoss: 0.5832 | Train Accuracy: 76.76
    | Val Loss 2.0538 | Val Accuracy: 49.94

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186 Epoch: 141 | NLLoss: 0.5513 | Train Accuracy: 78.00
187 Epoch: 142 | NLLoss: 0.5602 | Train Accuracy: 77.62
188 Epoch: 143 | NLLoss: 0.5390 | Train Accuracy: 78.57
189 Epoch: 144 | NLLoss: 0.5319 | Train Accuracy: 79.14
190 Epoch: 145 | NLLoss: 0.5352 | Train Accuracy: 78.57
191 Epoch: 146 | NLLoss: 0.5311 | Train Accuracy: 79.00
192 Epoch: 147 | NLLoss: 0.5341 | Train Accuracy: 79.00
193 Epoch: 148 | NLLoss: 0.5485 | Train Accuracy: 78.48
194 Epoch: 149 | NLLoss: 0.5093 | Train Accuracy: 79.95
195 Epoch: 150 | NLLoss: 0.4917 | Train Accuracy: 80.62
196 Validation ...
197 Epoch: 150 | NLLoss: 0.4917 | Train Accuracy: 80.62
    | Val Loss 2.1470 | Val Accuracy: 49.60
198 Epoch: 151 | NLLoss: 0.5006 | Train Accuracy: 81.00
199 Epoch: 152 | NLLoss: 0.5108 | Train Accuracy: 79.57
200 Epoch: 153 | NLLoss: 0.4858 | Train Accuracy: 80.76
201 Epoch: 154 | NLLoss: 0.4722 | Train Accuracy: 81.62
202 Epoch: 155 | NLLoss: 0.4620 | Train Accuracy: 81.76
203 Epoch: 156 | NLLoss: 0.4797 | Train Accuracy: 81.10
204 Epoch: 157 | NLLoss: 0.5155 | Train Accuracy: 79.19
205 Epoch: 158 | NLLoss: 0.9559 | Train Accuracy: 62.81
206 Epoch: 159 | NLLoss: 1.1584 | Train Accuracy: 58.95
207 Epoch: 160 | NLLoss: 0.9891 | Train Accuracy: 61.10
208 Validation ...
209 Epoch: 160 | NLLoss: 0.9891 | Train Accuracy: 61.10
    | Val Loss 2.0118 | Val Accuracy: 46.29
210 Epoch: 161 | NLLoss: 0.9219 | Train Accuracy: 63.90
211 Epoch: 162 | NLLoss: 0.7706 | Train Accuracy: 69.86
212 Epoch: 163 | NLLoss: 0.7033 | Train Accuracy: 73.14
213 Epoch: 164 | NLLoss: 0.6805 | Train Accuracy: 73.00
214 Epoch: 165 | NLLoss: 0.6147 | Train Accuracy: 76.05
215 Epoch: 166 | NLLoss: 0.5710 | Train Accuracy: 77.95
216 Epoch: 167 | NLLoss: 0.5415 | Train Accuracy: 78.90
217 Epoch: 168 | NLLoss: 0.5594 | Train Accuracy: 77.33
218 Epoch: 169 | NLLoss: 0.5703 | Train Accuracy: 77.29
219 Epoch: 170 | NLLoss: 0.5602 | Train Accuracy: 77.86
220 Validation ...
221 Epoch: 170 | NLLoss: 0.5602 | Train Accuracy: 77.86
    | Val Loss 1.9677 | Val Accuracy: 49.49
222 Epoch: 171 | NLLoss: 0.5587 | Train Accuracy: 77.57
223 Epoch: 172 | NLLoss: 0.5249 | Train Accuracy: 79.14

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224 Epoch: 173 | NLLoss: 0.5046 | Train Accuracy: 79.81
225 Epoch: 174 | NLLoss: 0.5190 | Train Accuracy: 79.52
226 Epoch: 175 | NLLoss: 0.5086 | Train Accuracy: 79.43
227 Epoch: 176 | NLLoss: 0.4893 | Train Accuracy: 81.00
228 Epoch: 177 | NLLoss: 0.4834 | Train Accuracy: 81.00
229 Epoch: 178 | NLLoss: 0.4738 | Train Accuracy: 81.43
230 Epoch: 179 | NLLoss: 0.4640 | Train Accuracy: 81.62
231 Epoch: 180 | NLLoss: 0.4723 | Train Accuracy: 80.86
232 Validation ...
233 Epoch: 180 | NLLoss: 0.4723 | Train Accuracy: 80.86
    | Val Loss 2.0940 | Val Accuracy: 48.69
234 Epoch: 181 | NLLoss: 0.4537 | Train Accuracy: 82.24
235 Epoch: 182 | NLLoss: 0.4665 | Train Accuracy: 81.57
236 Epoch: 183 | NLLoss: 0.4661 | Train Accuracy: 81.38
237 Epoch: 184 | NLLoss: 0.4911 | Train Accuracy: 81.10
238 Epoch: 185 | NLLoss: 0.5147 | Train Accuracy: 79.29
239 Epoch: 186 | NLLoss: 0.5748 | Train Accuracy: 76.76
240 Epoch: 187 | NLLoss: 0.6414 | Train Accuracy: 74.24
241 Epoch: 188 | NLLoss: 0.7780 | Train Accuracy: 69.48
242 Epoch: 189 | NLLoss: 0.6866 | Train Accuracy: 71.95
243 Epoch: 190 | NLLoss: 0.7083 | Train Accuracy: 71.62
244 Validation ...
245 Epoch: 190 | NLLoss: 0.7083 | Train Accuracy: 71.62
    | Val Loss 2.0361 | Val Accuracy: 48.23
246 Epoch: 191 | NLLoss: 0.6757 | Train Accuracy: 71.38
247 Epoch: 192 | NLLoss: 0.6450 | Train Accuracy: 72.19
248 Epoch: 193 | NLLoss: 0.5806 | Train Accuracy: 76.10
249 Epoch: 194 | NLLoss: 0.5150 | Train Accuracy: 78.57
250 Epoch: 195 | NLLoss: 0.4982 | Train Accuracy: 80.00
251 Epoch: 196 | NLLoss: 0.5299 | Train Accuracy: 78.86
252 Epoch: 197 | NLLoss: 0.5294 | Train Accuracy: 78.90
253 Epoch: 198 | NLLoss: 0.4995 | Train Accuracy: 79.33
254 Epoch: 199 | NLLoss: 0.4954 | Train Accuracy: 80.14
255 Epoch: 200 | NLLoss: 0.4641 | Train Accuracy: 81.81
256 Validation ...
257 Epoch: 200 | NLLoss: 0.4641 | Train Accuracy: 81.81
    | Val Loss 2.0159 | Val Accuracy: 50.86
258 Epoch: 201 | NLLoss: 0.4392 | Train Accuracy: 82.90
259 Epoch: 202 | NLLoss: 0.4344 | Train Accuracy: 82.86
260 Epoch: 203 | NLLoss: 0.4452 | Train Accuracy: 82.00
261 Epoch: 204 | NLLoss: 0.4247 | Train Accuracy: 82.90

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262 Epoch: 205 | NLLoss: 0.4333 | Train Accuracy: 82.19
263 Epoch: 206 | NLLoss: 0.4313 | Train Accuracy: 82.57
264 Epoch: 207 | NLLoss: 0.4282 | Train Accuracy: 82.43
265 Epoch: 208 | NLLoss: 0.4212 | Train Accuracy: 83.05
266 Epoch: 209 | NLLoss: 0.4158 | Train Accuracy: 83.00
267 Epoch: 210 | NLLoss: 0.5383 | Train Accuracy: 78.43
268 Validation ...
269 Epoch: 210 | NLLoss: 0.5383 | Train Accuracy: 78.43
    | Val Loss 2.1415 | Val Accuracy: 51.09
270 Epoch: 211 | NLLoss: 0.7371 | Train Accuracy: 70.62
271 Epoch: 212 | NLLoss: 0.6571 | Train Accuracy: 74.24
272 Epoch: 213 | NLLoss: 0.5226 | Train Accuracy: 79.19
273 Epoch: 214 | NLLoss: 0.4681 | Train Accuracy: 81.29
274 Epoch: 215 | NLLoss: 0.4603 | Train Accuracy: 81.57
275 Epoch: 216 | NLLoss: 0.4287 | Train Accuracy: 83.00
276 Epoch: 217 | NLLoss: 0.4395 | Train Accuracy: 82.71
277 Epoch: 218 | NLLoss: 0.4470 | Train Accuracy: 82.95
278 Epoch: 219 | NLLoss: 0.4927 | Train Accuracy: 80.33
279 Epoch: 220 | NLLoss: 0.5841 | Train Accuracy: 76.48
280 Validation ...
281 Epoch: 220 | NLLoss: 0.5841 | Train Accuracy: 76.48
    | Val Loss 2.0474 | Val Accuracy: 50.40
282 Epoch: 221 | NLLoss: 0.5228 | Train Accuracy: 79.48
283 Epoch: 222 | NLLoss: 0.4770 | Train Accuracy: 80.81
284 Epoch: 223 | NLLoss: 0.4503 | Train Accuracy: 82.43
285 Epoch: 224 | NLLoss: 0.5672 | Train Accuracy: 79.05
286 Epoch: 225 | NLLoss: 0.7228 | Train Accuracy: 71.38
287 Epoch: 226 | NLLoss: 0.8416 | Train Accuracy: 66.86
288 Epoch: 227 | NLLoss: 0.6675 | Train Accuracy: 74.05
289 Epoch: 228 | NLLoss: 0.5324 | Train Accuracy: 78.71
290 Epoch: 229 | NLLoss: 0.4900 | Train Accuracy: 81.05
291 Epoch: 230 | NLLoss: 0.4610 | Train Accuracy: 82.00
292 Validation ...
293 Epoch: 230 | NLLoss: 0.4610 | Train Accuracy: 82.00
    | Val Loss 1.9738 | Val Accuracy: 50.97
294 Epoch: 231 | NLLoss: 0.4336 | Train Accuracy: 83.29
295 Epoch: 232 | NLLoss: 0.4147 | Train Accuracy: 83.86
296 Epoch: 233 | NLLoss: 0.4049 | Train Accuracy: 84.10
297 Epoch: 234 | NLLoss: 0.4099 | Train Accuracy: 83.57
298 Epoch: 235 | NLLoss: 0.3991 | Train Accuracy: 84.48
299 Epoch: 236 | NLLoss: 0.3908 | Train Accuracy: 84.29

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300 Epoch: 237 | NLLoss: 0.3906 | Train Accuracy: 84.14
301 Epoch: 238 | NLLoss: 0.3906 | Train Accuracy: 84.19
302 Epoch: 239 | NLLoss: 0.3931 | Train Accuracy: 84.24
303 Epoch: 240 | NLLoss: 0.3798 | Train Accuracy: 84.52
304 Validation ...
305 Epoch: 240 | NLLoss: 0.3798 | Train Accuracy: 84.52
    | Val Loss 2.1191 | Val Accuracy: 50.29
306 Epoch: 241 | NLLoss: 0.3801 | Train Accuracy: 84.19
307 Epoch: 242 | NLLoss: 0.3839 | Train Accuracy: 84.43
308 Epoch: 243 | NLLoss: 0.3746 | Train Accuracy: 84.57
309 Epoch: 244 | NLLoss: 0.3770 | Train Accuracy: 84.86
310 Epoch: 245 | NLLoss: 0.3833 | Train Accuracy: 84.38
311 Epoch: 246 | NLLoss: 0.4290 | Train Accuracy: 83.00
312 Epoch: 247 | NLLoss: 0.4322 | Train Accuracy: 82.52
313 Epoch: 248 | NLLoss: 0.3987 | Train Accuracy: 84.43
314 Epoch: 249 | NLLoss: 0.4604 | Train Accuracy: 81.67
315 Epoch: 250 | NLLoss: 1.0148 | Train Accuracy: 64.71
316 Validation ...
317 Epoch: 250 | NLLoss: 1.0148 | Train Accuracy: 64.71
    | Val Loss 2.2251 | Val Accuracy: 45.14
318 Epoch: 251 | NLLoss: 1.0620 | Train Accuracy: 59.67
319 Epoch: 252 | NLLoss: 0.7583 | Train Accuracy: 69.86
320 Epoch: 253 | NLLoss: 0.5383 | Train Accuracy: 78.62
321 Epoch: 254 | NLLoss: 0.4883 | Train Accuracy: 80.67
322 Epoch: 255 | NLLoss: 0.4522 | Train Accuracy: 81.71
323 Epoch: 256 | NLLoss: 0.4182 | Train Accuracy: 83.38
324 Epoch: 257 | NLLoss: 0.4190 | Train Accuracy: 83.48
325 Epoch: 258 | NLLoss: 0.4013 | Train Accuracy: 83.81
326 Epoch: 259 | NLLoss: 0.3898 | Train Accuracy: 84.71
327 Epoch: 260 | NLLoss: 0.3923 | Train Accuracy: 84.33
328 Validation ...
329 Epoch: 260 | NLLoss: 0.3923 | Train Accuracy: 84.33
    | Val Loss 2.1744 | Val Accuracy: 48.91
330 Epoch: 261 | NLLoss: 0.3829 | Train Accuracy: 84.76
331 Epoch: 262 | NLLoss: 0.3883 | Train Accuracy: 84.76
332 Epoch: 263 | NLLoss: 0.3784 | Train Accuracy: 85.05
333 Epoch: 264 | NLLoss: 0.3763 | Train Accuracy: 85.00
334 Epoch: 265 | NLLoss: 0.3678 | Train Accuracy: 85.24
335 Epoch: 266 | NLLoss: 0.3664 | Train Accuracy: 85.00
336 Epoch: 267 | NLLoss: 0.3710 | Train Accuracy: 84.76
337 Epoch: 268 | NLLoss: 0.3581 | Train Accuracy: 85.33

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338 Epoch: 269 | NLLoss: 0.4026 | Train Accuracy: 84.14
339 Epoch: 270 | NLLoss: 0.5121 | Train Accuracy: 80.38
340 Validation ...
341 Epoch: 270 | NLLoss: 0.5121 | Train Accuracy: 80.38
    | Val Loss 2.1529 | Val Accuracy: 50.40
342 Epoch: 271 | NLLoss: 0.6581 | Train Accuracy: 74.52
343 Epoch: 272 | NLLoss: 0.6016 | Train Accuracy: 76.29
344 Epoch: 273 | NLLoss: 0.5275 | Train Accuracy: 79.43
345 Epoch: 274 | NLLoss: 0.4702 | Train Accuracy: 81.14
346 Epoch: 275 | NLLoss: 0.4488 | Train Accuracy: 82.14
347 Epoch: 276 | NLLoss: 0.4387 | Train Accuracy: 82.52
348 Epoch: 277 | NLLoss: 0.4472 | Train Accuracy: 82.67
349 Epoch: 278 | NLLoss: 0.3910 | Train Accuracy: 84.48
350 Epoch: 279 | NLLoss: 0.4739 | Train Accuracy: 81.00
351 Epoch: 280 | NLLoss: 0.4527 | Train Accuracy: 82.14
352 Validation ...
353 Epoch: 280 | NLLoss: 0.4527 | Train Accuracy: 82.14
    | Val Loss 2.2098 | Val Accuracy: 50.51
354 Epoch: 281 | NLLoss: 0.4517 | Train Accuracy: 81.95
355 Epoch: 282 | NLLoss: 0.4970 | Train Accuracy: 79.81
356 Epoch: 283 | NLLoss: 0.4610 | Train Accuracy: 81.00
357 Epoch: 284 | NLLoss: 0.4219 | Train Accuracy: 83.81
358 Epoch: 285 | NLLoss: 0.3854 | Train Accuracy: 84.10
359 Epoch: 286 | NLLoss: 0.4036 | Train Accuracy: 83.86
360 Epoch: 287 | NLLoss: 0.4064 | Train Accuracy: 84.05
361 Epoch: 288 | NLLoss: 0.4208 | Train Accuracy: 83.76
362 Epoch: 289 | NLLoss: 0.3870 | Train Accuracy: 84.29
363 Epoch: 290 | NLLoss: 0.3781 | Train Accuracy: 85.57
364 Validation ...
365 Epoch: 290 | NLLoss: 0.3781 | Train Accuracy: 85.57
    | Val Loss 2.1910 | Val Accuracy: 51.43
366 Epoch: 291 | NLLoss: 0.3630 | Train Accuracy: 85.14
367 Epoch: 292 | NLLoss: 0.3516 | Train Accuracy: 85.90
368 Epoch: 293 | NLLoss: 0.3412 | Train Accuracy: 86.33
369 Epoch: 294 | NLLoss: 0.3436 | Train Accuracy: 86.38
370 Epoch: 295 | NLLoss: 0.3433 | Train Accuracy: 86.00
371 Epoch: 296 | NLLoss: 0.3516 | Train Accuracy: 86.14
372 Epoch: 297 | NLLoss: 0.3512 | Train Accuracy: 85.81
373 Epoch: 298 | NLLoss: 0.3361 | Train Accuracy: 86.86
374 Epoch: 299 | NLLoss: 0.3317 | Train Accuracy: 87.00
375 Epoch: 300 | NLLoss: 0.3632 | Train Accuracy: 85.76
```

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376 Validation ...
377 Epoch: 300 | NLLoss: 0.3632 | Train Accuracy: 85.76
    | Val Loss 2.1948 | Val Accuracy: 49.60
378 Epoch: 301 | NLLoss: 0.3541 | Train Accuracy: 86.05
379 Epoch: 302 | NLLoss: 0.3376 | Train Accuracy: 86.62
380 Epoch: 303 | NLLoss: 0.3366 | Train Accuracy: 86.76
381 Epoch: 304 | NLLoss: 0.3316 | Train Accuracy: 87.05
382 Epoch: 305 | NLLoss: 0.3299 | Train Accuracy: 87.00
383 Epoch: 306 | NLLoss: 0.3267 | Train Accuracy: 87.05
384 Epoch: 307 | NLLoss: 0.3373 | Train Accuracy: 86.38
385 Epoch: 308 | NLLoss: 0.3423 | Train Accuracy: 86.00
386 Epoch: 309 | NLLoss: 0.3440 | Train Accuracy: 86.43
387 Epoch: 310 | NLLoss: 0.3330 | Train Accuracy: 86.81
388 Validation ...
389 Epoch: 310 | NLLoss: 0.3330 | Train Accuracy: 86.81
    | Val Loss 2.3669 | Val Accuracy: 51.09
390 Epoch: 311 | NLLoss: 0.3271 | Train Accuracy: 87.24
391 Epoch: 312 | NLLoss: 0.3207 | Train Accuracy: 87.38
392 Epoch: 313 | NLLoss: 0.3502 | Train Accuracy: 86.67
393 Epoch: 314 | NLLoss: 0.5609 | Train Accuracy: 78.10
394 Epoch: 315 | NLLoss: 1.1554 | Train Accuracy: 61.62
395 Epoch: 316 | NLLoss: 1.5642 | Train Accuracy: 47.38
396 Epoch: 317 | NLLoss: 1.0237 | Train Accuracy: 59.71
397 Epoch: 318 | NLLoss: 0.8148 | Train Accuracy: 68.24
398 Epoch: 319 | NLLoss: 0.6373 | Train Accuracy: 74.33
399 Epoch: 320 | NLLoss: 0.6002 | Train Accuracy: 75.67
400 Validation ...
401 Epoch: 320 | NLLoss: 0.6002 | Train Accuracy: 75.67
    | Val Loss 1.9239 | Val Accuracy: 49.49
402 Epoch: 321 | NLLoss: 0.5614 | Train Accuracy: 77.43
403 Epoch: 322 | NLLoss: 0.6216 | Train Accuracy: 76.05
404 Epoch: 323 | NLLoss: 0.6170 | Train Accuracy: 75.76
405 Epoch: 324 | NLLoss: 0.4994 | Train Accuracy: 80.95
406 Epoch: 325 | NLLoss: 0.4643 | Train Accuracy: 81.62
407 Epoch: 326 | NLLoss: 0.4225 | Train Accuracy: 83.86
408 Epoch: 327 | NLLoss: 0.3841 | Train Accuracy: 84.90
409 Epoch: 328 | NLLoss: 0.4221 | Train Accuracy: 83.10
410 Epoch: 329 | NLLoss: 0.4385 | Train Accuracy: 82.81
411 Epoch: 330 | NLLoss: 0.4337 | Train Accuracy: 83.24
412 Validation ...
413 Epoch: 330 | NLLoss: 0.4337 | Train Accuracy: 83.24
```

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413 | Val Loss 2.1965 | Val Accuracy: 49.60
414 Epoch: 331 | NLLoss: 0.4018 | Train Accuracy: 84.76
415 Epoch: 332 | NLLoss: 0.4246 | Train Accuracy: 83.52
416 Epoch: 333 | NLLoss: 0.4286 | Train Accuracy: 82.38
417 Epoch: 334 | NLLoss: 0.3896 | Train Accuracy: 84.81
418 Epoch: 335 | NLLoss: 0.3579 | Train Accuracy: 85.90
419 Epoch: 336 | NLLoss: 0.3493 | Train Accuracy: 86.29
420 Epoch: 337 | NLLoss: 0.3484 | Train Accuracy: 86.48
421 Epoch: 338 | NLLoss: 0.3396 | Train Accuracy: 86.48
422 Epoch: 339 | NLLoss: 0.3295 | Train Accuracy: 86.95
423 Epoch: 340 | NLLoss: 0.3266 | Train Accuracy: 87.24
424 Validation ...
425 Epoch: 340 | NLLoss: 0.3266 | Train Accuracy: 87.24
    | Val Loss 2.2678 | Val Accuracy: 48.69
426 Epoch: 341 | NLLoss: 0.3175 | Train Accuracy: 87.43
427 Epoch: 342 | NLLoss: 0.3224 | Train Accuracy: 87.00
428 Epoch: 343 | NLLoss: 0.3158 | Train Accuracy: 87.71
429 Epoch: 344 | NLLoss: 0.3250 | Train Accuracy: 87.90
430 Epoch: 345 | NLLoss: 0.3182 | Train Accuracy: 88.14
431 Epoch: 346 | NLLoss: 0.3070 | Train Accuracy: 87.81
432 Epoch: 347 | NLLoss: 0.3059 | Train Accuracy: 88.67
433 Epoch: 348 | NLLoss: 0.3054 | Train Accuracy: 88.14
434 Epoch: 349 | NLLoss: 0.3080 | Train Accuracy: 88.14
435 Epoch: 350 | NLLoss: 0.3325 | Train Accuracy: 86.86
436 Validation ...
437 Epoch: 350 | NLLoss: 0.3325 | Train Accuracy: 86.86
    | Val Loss 2.2957 | Val Accuracy: 50.63
438 Epoch: 351 | NLLoss: 0.4571 | Train Accuracy: 81.29
439 Epoch: 352 | NLLoss: 0.6310 | Train Accuracy: 74.62
440 Epoch: 353 | NLLoss: 0.6034 | Train Accuracy: 77.48
441 Epoch: 354 | NLLoss: 0.5726 | Train Accuracy: 77.10
442 Epoch: 355 | NLLoss: 0.5306 | Train Accuracy: 78.43
443 Epoch: 356 | NLLoss: 0.4683 | Train Accuracy: 81.29
444 Epoch: 357 | NLLoss: 0.5027 | Train Accuracy: 80.24
445 Epoch: 358 | NLLoss: 0.4266 | Train Accuracy: 83.24
446 Epoch: 359 | NLLoss: 0.3568 | Train Accuracy: 86.48
447 Epoch: 360 | NLLoss: 0.3387 | Train Accuracy: 86.95
448 Validation ...
449 Epoch: 360 | NLLoss: 0.3387 | Train Accuracy: 86.95
    | Val Loss 2.1874 | Val Accuracy: 49.71
450 Epoch: 361 | NLLoss: 0.3298 | Train Accuracy: 87.76

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451 Epoch: 362 | NLLoss: 0.3201 | Train Accuracy: 87.43
452 Epoch: 363 | NLLoss: 0.3064 | Train Accuracy: 88.33
453 Epoch: 364 | NLLoss: 0.3003 | Train Accuracy: 88.76
454 Epoch: 365 | NLLoss: 0.3023 | Train Accuracy: 88.52
455 Epoch: 366 | NLLoss: 0.2954 | Train Accuracy: 88.71
456 Epoch: 367 | NLLoss: 0.2946 | Train Accuracy: 88.38
457 Epoch: 368 | NLLoss: 0.2891 | Train Accuracy: 88.57
458 Epoch: 369 | NLLoss: 0.2859 | Train Accuracy: 89.10
459 Epoch: 370 | NLLoss: 0.2960 | Train Accuracy: 88.76
460 Validation ...
461 Epoch: 370 | NLLoss: 0.2960 | Train Accuracy: 88.76
    | Val Loss 2.4290 | Val Accuracy: 49.94
462 Epoch: 371 | NLLoss: 0.3517 | Train Accuracy: 85.67
463 Epoch: 372 | NLLoss: 0.3092 | Train Accuracy: 87.38
464 Epoch: 373 | NLLoss: 0.3818 | Train Accuracy: 84.71
465 Epoch: 374 | NLLoss: 0.4819 | Train Accuracy: 80.71
466 Epoch: 375 | NLLoss: 0.4356 | Train Accuracy: 83.48
467 Epoch: 376 | NLLoss: 0.3640 | Train Accuracy: 85.43
468 Epoch: 377 | NLLoss: 0.3770 | Train Accuracy: 85.48
469 Epoch: 378 | NLLoss: 0.3512 | Train Accuracy: 86.19
470 Epoch: 379 | NLLoss: 0.3306 | Train Accuracy: 86.76
471 Epoch: 380 | NLLoss: 0.4445 | Train Accuracy: 82.57
472 Validation ...
473 Epoch: 380 | NLLoss: 0.4445 | Train Accuracy: 82.57
    | Val Loss 2.3695 | Val Accuracy: 48.91
474 Epoch: 381 | NLLoss: 0.4043 | Train Accuracy: 83.43
475 Epoch: 382 | NLLoss: 0.3641 | Train Accuracy: 86.10
476 Epoch: 383 | NLLoss: 0.3890 | Train Accuracy: 85.14
477 Epoch: 384 | NLLoss: 0.3537 | Train Accuracy: 85.90
478 Epoch: 385 | NLLoss: 0.3506 | Train Accuracy: 85.48
479 Epoch: 386 | NLLoss: 0.3422 | Train Accuracy: 87.14
480 Epoch: 387 | NLLoss: 0.3329 | Train Accuracy: 87.14
481 Epoch: 388 | NLLoss: 0.3180 | Train Accuracy: 87.05
482 Epoch: 389 | NLLoss: 0.3008 | Train Accuracy: 88.14
483 Epoch: 390 | NLLoss: 0.2986 | Train Accuracy: 88.05
484 Validation ...
485 Epoch: 390 | NLLoss: 0.2986 | Train Accuracy: 88.05
    | Val Loss 2.3211 | Val Accuracy: 51.54
486 Epoch: 391 | NLLoss: 0.2905 | Train Accuracy: 88.29
487 Epoch: 392 | NLLoss: 0.2928 | Train Accuracy: 88.00
488 Epoch: 393 | NLLoss: 0.3069 | Train Accuracy: 87.00

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489 Epoch: 394 | NLLoss: 0.2911 | Train Accuracy: 88.24
490 Epoch: 395 | NLLoss: 0.2821 | Train Accuracy: 88.71
491 Epoch: 396 | NLLoss: 0.2751 | Train Accuracy: 89.14
492 Epoch: 397 | NLLoss: 0.2786 | Train Accuracy: 89.43
493 Epoch: 398 | NLLoss: 0.2920 | Train Accuracy: 88.29
494 Epoch: 399 | NLLoss: 0.2723 | Train Accuracy: 89.24
495
496 Process finished with exit code 0
497
```